

An annotation-free structural readout of CDS and UTR sequence regimes in the human genome

Jieqi Liu

Independent researcher

Correspondence: jackey.l.gene@outlook.com

June 2026

Abstract

Genomic sequence analysis typically begins with annotation: gene models, functional domains, expression data, or evolutionary conservation. We introduce **SHP (Saussurean Hash Projection)**, a lightweight, zero-training structural readout for nucleotide sequences. Feature computation is annotation-free after CDS/UTR sequence extraction: Ensembl transcript annotations are used to define sequence regions, but no functional labels, expression data, conservation scores, or disease annotations enter the SHP calculation. SHP encodes each sequence window through two complementary views of the same 3-mer stream—a *chroma* axis (which 3-mers are present) and a *rhythm* axis (which 3-mer transitions occur)—into the same 64-dimensional hash space, then measures structural tension via the Jaccard distance between them. The instrument is calibrated against a fair IID baseline ($k=4$, $\theta_0=0.0999$).

Applied to the full human protein-coding genome (19,491 genes, 224,518 transcript isoforms, Ensembl release 115), SHP produces an 8-dimensional per-gene structural feature vector (fixed_wit, tail_energy, skew, kurt for both CDS and UTR) plus derived

gradients, without consulting any external functional labels. We report the following findings: (1) the CDS structural quiescence rate is 48.8% genome-wide, with regime-specific signatures—MHC class I genes show 80% CDS quiescence, while neural genes show only 20%; (2) a UTR/CDS structural gradient distinguishes functional regimes, with KRTAP genes as the only CDS-led regime (85.7% UTR quiescence); (3) SHP vectors carry functional information: nearest-centroid classification achieves 30–40% per-category recall for olfactory receptors, HLA genes, and keratins from 2D SHP features alone; (4) this clustering is not a GC-content proxy—zinc-finger genes (GC=47%) and transcription factors (GC=63%) are separated by 16 percentage points of GC yet cluster together in SHP space ($d=0.004$). The method requires no training, no model weights, and no external functional databases beyond the input sequence files. We provide a calibrated, annotation-free structural spectroscopy tool for genomic sequence screening.

Introduction

Modern genomics produces vast catalogs of sequences — transcript isoforms, variant calls, regulatory regions — but experimental characterization remains expensive. Computational triage typically relies on sequence homology, functional domain databases, expression correlation, codon-usage summaries, or evolutionary conservation scores (for example CAI (Sharp and Li, 1987), GO/InterPro (Gene Ontology Consortium, 2023; Blum et al., 2025), or phyloP-style conservation scores (Pollard et al., 2010)). Each of these carries prior assumptions about what constitutes functional importance.

Here we introduce an orthogonal approach: direct structural spectroscopy of the nucleotide symbol stream itself. The method, SHP (Saussurean Hash Projection), asks a single question of any DNA sequence: *does the local 3-mer composition (chroma) and the local 3-mer transition pattern (rhythm) exhibit structural tension beyond what a random sequence would produce?*

The name is a compact mnemonic rather than a biological assumption. Its two axes loosely echo the paradigmatic/syntagmatic distinction in structural linguistics (Saussure, 1983): what symbols are locally available, and how symbols are chained. The algorithm itself requires no linguistic theory; it only compares binary k-mer presence and binary k-mer transition activations in a shared hash space.

The present paper treats SHP as a standalone calibrated screening instrument for genomic sequences. No prior knowledge of the broader symbolic-processing framework in which the dual-axis encoding was developed is required to use or interpret the results.

The core measurement is straightforward. A sliding window of 128 nucleotides is projected into a 64-dimensional hash space twice: once encoding which 3-mers appear (chroma), and once encoding which 3-mer-to-3-mer transitions occur (rhythm). The Jaccard distance $1 - |C_t \cap R_t| / |C_t \cup R_t|$ (Jaccard, 1901) between these two binary activation patterns—the *cross-harm* h_t —measures the structural tension between the paradigmatic and syntagmatic axes of the local sequence grammar. A structural event is registered when the cross-harm displacement $d_t = |h_t - h_{t-1}|$ exceeds the 99th percentile of fluctuations observed in a maximally unstructured (fair IID) stream of the same alphabet.

The instrument was calibrated on synthetic symbol streams spanning GC content from 40–70%, CpG suppression, and codon periodicity; the resolution threshold ($\theta_0 = 0.0999$

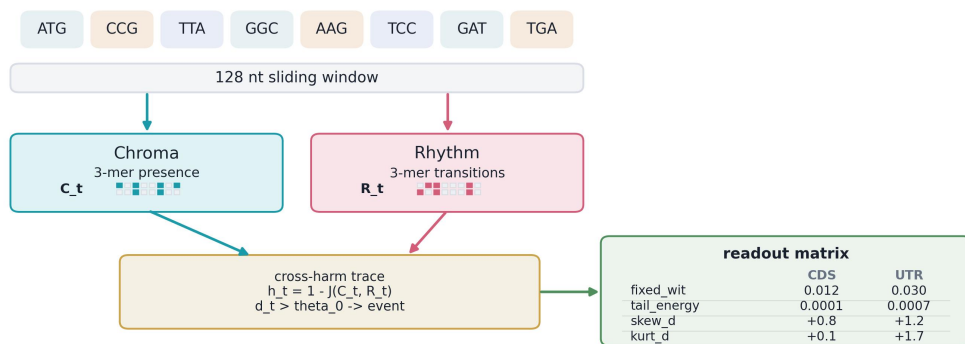
for 128-nt windows) proved stable across these controls. We then applied it to all human protein-coding genes with measurable CDS and UTR (Ensembl release 115, 19,491 genes), producing per-gene structural vectors without using any gene ontology terms, disease labels, expression data, or conservation scores.

The resulting SHP matrix reveals regime-level structural signatures, carries functional information orthogonal to GC content, and identifies structurally extreme genes and isoforms that no external annotation would have flagged. We present the method, the calibration, the genome-scale results, and a practical tool for annotation-free genomic sequence screening.

Results

Figure 1. SHP turns one nucleotide stream into two structural views

Chroma asks what 3-mers are present. Rhythm asks how 3-mers transition. Cross-harm measures their mismatch.



Chroma asks which 3-mers are present; rhythm asks how 3-mers transition. Cross-harm measures their mismatch.

SHP Calibration and Measurement Framework

SHP operates on any discrete symbol stream with alphabet size $k \geq 4$. For DNA ($k=4$), a sliding window of $W=128$ nucleotides is analyzed at $n=3$ (trinucleotide) granularity with $D=64$ hash buckets, the combinatorial saturation point $k^3=D$. Within each window, chroma and rhythm vectors are computed as binary hash activations (presence/absence, not frequency-weighted). Cross-harm $h_t=1-J(C_t, R_t)$ is computed per window, where J denotes the Jaccard similarity of the two hash-activation sets. Displacement $d_t=|h_t-h_{t-1}|$ between consecutive windows forms the primary signal trace.

The instrument is calibrated against a fair IID baseline: 10,000-symbol streams with $k=4$ symbols drawn independently with uniform probability, repeated across 20 random seeds. The 99th percentile of cross-harm displacement under this baseline defines the resolution threshold $\theta_0=0.0999 \pm 0.0067$ (mean $\pm$ 1 SD across seeds).

From this calibration, we define two primary SHP metrics per sequence:

- **fixed_wit** = $\text{COUNT}(d_t > \theta_0) / N_{\text{transitions}}$ — the structural event rate, measuring how frequently the local 3-mer grammar undergoes reorganization beyond the random-fluctuation baseline.
- **tail_energy** = $\overline{\max(0, d_t - \theta_0)}$ — the structural event intensity, measuring the magnitude of excess displacement above baseline.

A third, derived metric—the distribution shape (skew, kurt) of $\{d_t\}$ —captures the structural phase of the sequence (periodic, random, Markov-biased, block-structured).

We validated θ_0 against DNA-specific composition constraints: GC content from 40–70%, CpG-suppressed dinucleotide Markov chains, and codon-periodic synthetic CDS all produce θ_0 within 5% of the fair-IID reference (no pairwise t -test significant at

Bonferroni-corrected $\alpha=0.01$). The threshold is stable for the tested DNA parameter setting ($k=4, n=3, D=64, W=128$).

A critical methodological finding: when each stream uses its own P99 as threshold (self-thresholding), all streams produce a fixed_wit of approximately 0.007. This apparent constancy is a statistical artifact of percentile-based thresholding. Under the fixed θ_0 , fixed_wit spans a 10-fold range across stream types. All results reported below use the fixed $\theta_0=0.0999$.

Genome-Wide SHP Matrix

We processed all 19,491 human protein-coding genes with measurable CDS (≥ 128 nt) from Ensembl release 115, plus 16,572 genes with measurable UTR. For each gene, SHP metrics were computed for the longest CDS transcript and its matching UTR (extracted by transcript-ID alignment of cDNA minus CDS). An additional 224,518 individual transcript isoforms were processed for isoform-level analysis.

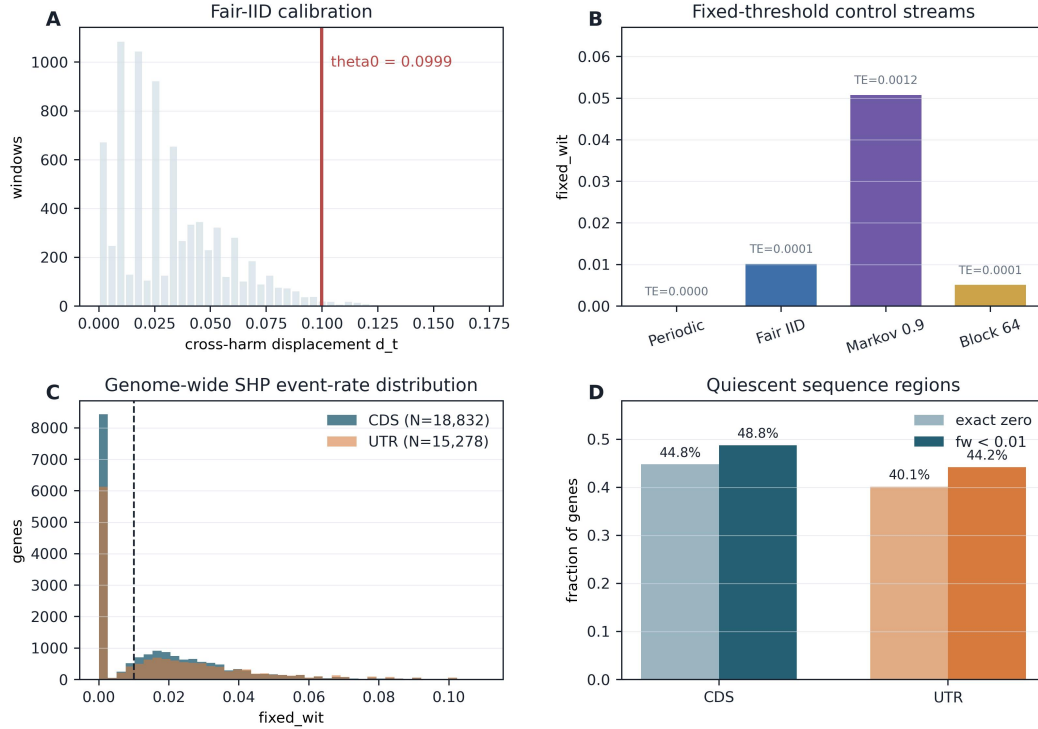
After applying a minimum-window filter ($n_{\text{windows}} \geq 10$, removing 659 genes with insufficient window coverage), the genome-wide distribution of SHP metrics is:

Table 1. Genome-wide SHP structural event rates in CDS and UTR regions.

Metric	CDS ($n_w \geq 10$)	UTR ($n_w \geq 10$)
Genes	18,832	15,278
Mean fixed_wit	0.0165	0.0201
Exactly zero (fw=0)	44.8%	40.1%
Quiescent (fw<0.01)	48.8%	44.2%
Mean gradient (UTR-CDS), both-region genes ($N=14,996$)	—	+0.0035

Approximately 45% of protein-coding genes show exactly zero structural events in their CDS (48.8% below the 0.01 quiescence threshold). UTRs are consistently more structurally active than CDSs in the both-region subset (mean gradient +0.0035, N=14,996 genes with both CDS and UTR passing the window filter).

Figure 2. Calibration and genome-wide structural event rates



(A) Fair-IID displacement distribution defining $\theta_0=0.0999$. (B) Fixed-threshold readout for control streams. (C) Genome-wide CDS and UTR fixed_wit distributions. (D) Exact-zero and quiescent fractions.

Regime-Level Structural Signatures

We examined five biological regimes with expanded gene sets ($N \geq 20$ per regime), selected to represent distinct functional categories. Gene lists are documented in the Methods and provided in the supplementary material.

Table 2. Regime-level CDS/UTR SHP signatures across representative gene categories.

Regime	N	CDS fw	UTR fw	CDS fw<0.01	UTR fw<0.01
HOX (developmental)	39	0.0104	0.0206	69.2%	43.6%
MHC (immune diversity)	20	0.0096	0.0303	80.0%	44.4%
BRAIN (neural)	25	0.0214	0.0294	20.0%	36.4%
AD (maintenance)	20	0.0115	0.0200	55.0%	52.6%
KRTAP (production)	75	0.0341	0.0110	58.7%	85.7%

MHC genes show the highest CDS quiescence of any regime (80% fw<0.01). Class I and class II HLA genes have structurally stable coding templates, with elevated structural activity in UTR regions (UTR fw=0.0303, highest of any regime). Whether this reflects selection for stable antigen-presentation domains coupled with regulated expression is a hypothesis for further investigation.

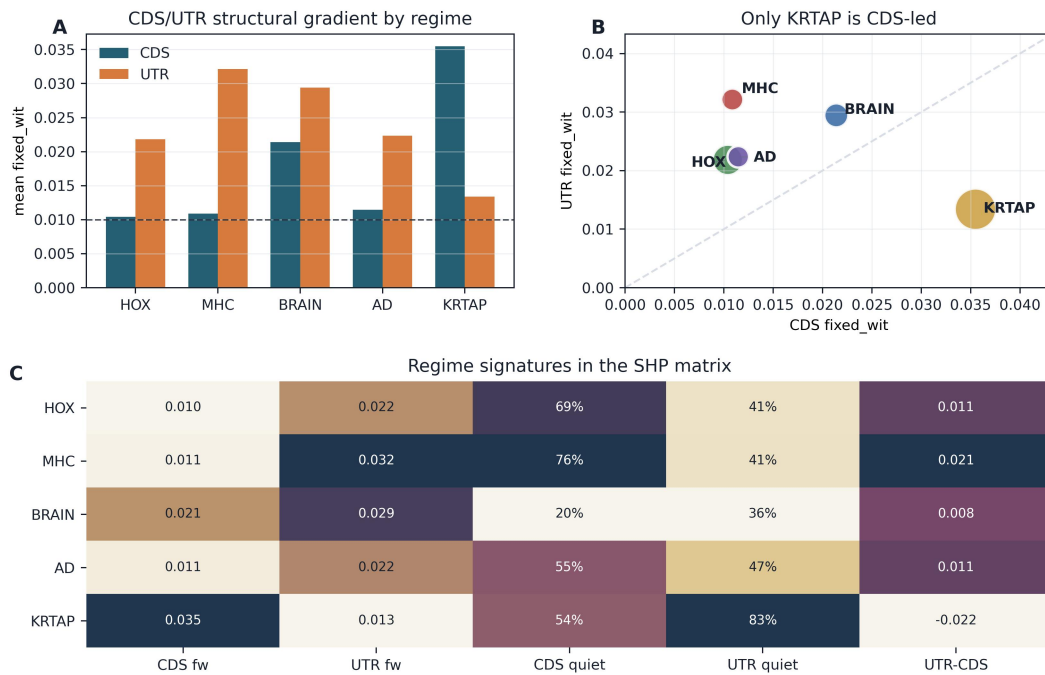
Neural genes show the lowest CDS quiescence (20% fw<0.01). Only one in five brain-expressed genes has a quiescent CDS template. Neural genes carry structural grammar throughout their coding regions — a phenotype consistent with extensive alternative splicing documented in neural tissue, though SHP does not directly measure splicing.

KRTAP genes are the only CDS-led regime. Production-type keratin-associated proteins show the highest UTR quiescence (85.7% fw<0.01) and a strongly negative UTR-CDS gradient (-0.027). Their structural grammar is concentrated in the coding region.

AD maintenance genes show the highest UTR quiescence among non-production regimes (52.6% fw<0.01). This observation is reported as an SHP phenotype; whether it reflects regulatory grammar changes associated with maintenance failure requires independent biological validation.

Within the expanded PRB/KRTAP production-gene family (N=100), we identified four structural phases across the CDS fixed_wit vs UTR fixed_wit plane: QUIET (81% of genes, both axes inactive), UTR-CONTROLLED (11%, CDS=0 + UTR>0.01), CDS-ACTIVE (5%), and DUAL-ACTIVE (3%). The UTR-CONTROLLED phase, exemplified by PRB3 (UTR fixed_wit=0.333, 39x baseline), was validated by dinucleotide-preserving shuffle (null fixed_wit=0.004, ratio=20x), confirming the signal is transition-driven rather than a composition artifact.

Figure 3. Biological regimes occupy distinct CDS/UTR structural states



(A) Mean CDS and UTR fixed_wit by regime. (B) Regime centroids in CDS/UTR structural space. (C) Min-max normalized heat map of regime signatures.

SHP Vectors Carry Functional Information

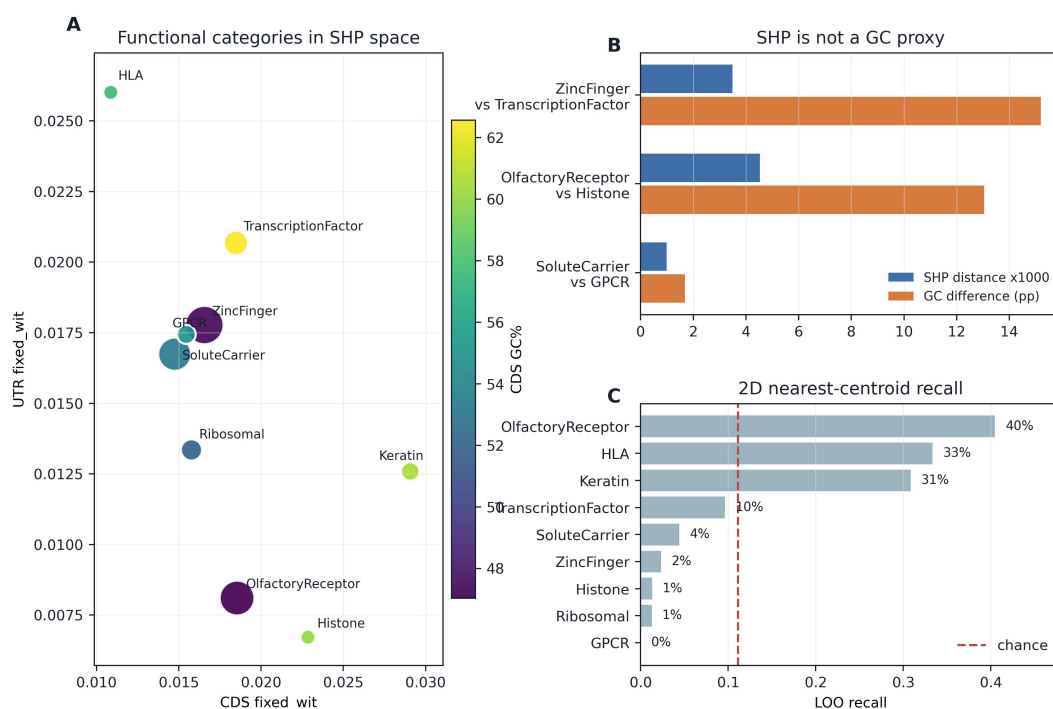
We tested whether SHP structural vectors carry functional signal by assigning broad categories based on gene symbol prefixes (e.g., all ZNF* genes as “ZincFinger”, all

SLC* as “SoluteCarrier”) — a deliberately crude classification using zero external annotation databases. Category labels were used only for validation, not as input features. Using only 2-dimensional SHP vectors (CDS fixed_wit, UTR fixed_wit), nearest-centroid leave-one-out classification achieved 13.2% overall accuracy vs. 11.1% random baseline (9 categories) across categories (N=2,034 labeled genes). Per-category recall: OlfactoryReceptor 40.5%, HLA 33.3%, Keratin 30.8% — three categories at 3–4x chance level from pure 3-mer structural grammar alone. GPCR and Histone categories showed near-chance performance in this low-dimensional baseline.

Crucially, SHP clustering is not a GC-content proxy. ZincFinger genes (GC=47.4%) and TranscriptionFactors (GC=62.6%) are separated by 16 percentage points of GC — yet in unscaled SHP fixed_wit space, their centroids are nearly overlapping (dist_SHP=0.004). Both are DNA-binding proteins. In 8 of 9 discriminative category pairs, SHP clustering disagreed with GC+length clustering. OlfactoryReceptors (GC=47.0%) and Histones (GC=60.1%) are separated by 13pp of GC yet cluster together in SHP space. Bootstrap validation (500 resamples, equal-N=12 subsampling, 36 pairwise comparisons) confirmed all 36 category pairs have 95% CI lower bounds excluding zero.

A label permutation test (200 shuffles) confirmed the observed 13.2% classification accuracy exceeds the null distribution (mean null accuracy $6.9\% \pm 2.5\%$, $Z=2.5$, $p<0.01$). The modest absolute accuracy reflects the deliberately low-dimensional feature space (2 of 8 available SHP dimensions).

Figure 4. SHP carries functional signal orthogonal to simple composition



(A) Functional categories in SHP space. (B) SHP distance vs GC difference for selected pairs. (C) Leave-one-out nearest-centroid recall.

Isoform Structural Diversity

Within-gene isoform SHP dispersion was computed for 17,111 genes with ≥ 2 isoforms. Mean within-gene fixed_wit range = 0.037. Top genes by isoform structural diversity: PPFIBP2 (20 isoforms, range 0.6), G3BP2 (73 isoforms, range 0.4), PUM1 (220 isoforms, range 0.4). Notably, 27.5% of multi-isoform genes have ALL isoforms at fixed_wit=0 — splicing produces zero structural grammar variation among their transcripts.

Chromosome-Scale Distribution

SHP metrics were mapped to genomic coordinates using GENCODE v49 gene annotations. All autosomes occupy a narrow band of mean CDS fixed_wit (0.0148–0.0182). Acrocentric chromosomes 21 and 22 show elevated short-arm/long-arm fixed_wit ratios (1.52 and 1.42 respectively). Chromosome Y is the most boundary-rich

(16 structural boundaries on 39 genes), consistent with palindromic repeat structure concentrating cross-harm reorganization at repeat boundaries.

Telomere proximity shows no effect on SHP metrics: CDS fixed_wit is flat across all 10 position bins from telomere to centromere (mean 0.016 +/- 0.001). This null result confirms SHP reads sequence-intrinsic 3-mer grammar — telomere position effects operate on expression state, not on CDS sequence structure.

Discussion

What SHP Measures

SHP occupies an unusual position in the genomic analysis toolkit. It is not a composition measure (GC content, codon adaptation index), not a conservation measure (phyloP, dN/dS), and not a functional predictor (GO term enrichment, domain detection). It measures *structural grammar* — the tension between which 3-mers are present (the paradigmatic axis) and how they transition (the syntagmatic axis) in a local sequence window. A sequence with high structural tension has a 3-mer composition that does not imply its 3-mer transition pattern; a sequence with zero structural events has these two axes in stable equilibrium.

This makes SHP complementary to existing tools. A gene can be highly conserved (strong phyloP score) yet have zero SHP structural events (stable 3-mer grammar). Conversely, a gene can show extreme SHP structural activity yet have no known disease association. The annotation-free nature of SHP means it can flag structurally unusual

sequences that external annotations have not yet characterized — a screening function orthogonal to knowledge-based prioritization.

Limitations

Window size sensitivity. The standard $W=128$ nt window (~42 codons) limits UTR analysis. Short UTRs (<128 nt) are excluded from the current matrix. A $W=64$ sensitivity mode is calibrated ($\theta_0=0.1214$) but was not applied genome-wide in this study.

Two-dimensional baseline. The functional clustering results use only CDS `fixed_wit` and UTR `fixed_wit`. The full 8-dimensional primary SHP vector (`fixed_wit`, `tail_energy`, skew, and kurt for both CDS and UTR) plus derived gradients may improve classification accuracy. The 13.2% overall accuracy reported here should be interpreted as a conservative lower bound.

Gene symbol classification. Functional categories were assigned by gene symbol prefix — a deliberately crude method. Formal GO/InterPro enrichment analysis is needed to validate the functional signal with standard ontologies.

Cross-species validation. The current matrix is human-only. The CDS quiescence rate of 48.8% and regime signatures need to be tested in other species. Preliminary primate analysis (chimpanzee, gorilla, orangutan) suggests the production-gene CDS quiescence rate is conserved (77–82%), but this requires raw genomic DNA rather than CDS annotation pipelines for definitive cross-species comparison.

Practical Utility

SHP is intentionally lightweight. The entire genome scan (19,491 genes, 224,518 isoforms) completes in approximately 1.7 hours (~6,200 CPU-seconds) on a single core. No GPU, no model weights, and no external functional databases are required once the

input sequence files are available. The method is implementable in ~200 lines of Python using only the standard library (hashlib, random, math).

For a bioinformatics user, SHP provides an orthogonal screening coordinate: given a gene list, SHP can rank genes by structural quiescence (CDS fixed_wit), regulatory structure (UTR fixed_wit), structural gradient (UTR-CDS difference), or isoform structural diversity (within-gene fixed_wit range). These rankings require no prior knowledge about gene function and can be computed for any organism with a CDS FASTA.

Methods

SHP Encoding

For a nucleotide sequence of length L , a sliding window of $W=128$ nucleotides is applied with stride $\max(1, W/5)$. For each window at position t :

Chroma vector: each overlapping 3-mer in the window is hashed via MD5, and the corresponding bucket in a $D=64$ binary vector is set to 1. Formally, $C_t[i]=1$ if any 3-mer in window t hashes to bucket i , else 0.

Rhythm vector: each adjacent 3-mer pair (3-mer _{t} , 3-mer _{$t+1$}) is hashed with a distinct label prefix, and the corresponding bucket in the same $D=64$ hash space is set to 1. Formally, $R_t[i]=1$ if any 3-mer transition in window t hashes to bucket i , else 0.

Cross-harm: $h_t = 1 - |C_t \cap R_t| / |C_t \cup R_t|$, the Jaccard distance between the two hash-activation sets.

Displacement: $d_t = |h_t - h_{t-1}|$ for $t \geq 2$; $d_1 = 0$.

Calibration

A fair IID stream of $T=10,000$ symbols from the $k=4$ alphabet $\{A, C, G, T\}$ with uniform probability $p=0.25$ per symbol is generated. d_t values are computed across sliding windows. The 99th percentile of the empirical d_t distribution, averaged across 20 independent random seeds, defines $\theta_0=0.0999\pm0.0067$.

For $W=64$ (short-UTR sensitivity mode), the same procedure yields $\theta_0=0.1214\pm0.0082$.

Biological Data

Human CDS sequences were obtained from Ensembl release 115 (Dyer et al., 2025) (Homo_sapiens.GRCh38.cds.all.fa.gz). Human cDNA sequences from the same release (Homo_sapiens.GRCh38.cdna.all.fa.gz) were used for UTR extraction. CDS and cDNA were matched by transcript ID (ENST*); UTR was extracted as the concatenation of 5' and 3' cDNA regions flanking the CDS match. For genes with multiple transcripts, the longest CDS with a transcript-matched cDNA was used for gene-level analysis.

Gene Regime Definitions

- **HOX:** 39 genes (HOXA, HOXB, HOXC, HOXD)
- **MHC:** 20 genes (HLA-A, -B, -C, -E, -F, -G, -H, -J, -L; HLA-DMA, -DMB, -DOA, -DOB, -DPA1, -DPB1, -DQA1, -DQA2, -DQB1, -DRA, -DRB1, -DRB3, -DRB4, -DRB5)
- **BRAIN:** 25 genes (FOXP1/2, PAX6, NEUROD1/2/4/6, TBR1, DLG4, SYN1/2/3, GABRA1-6, GRIN1/2A/2B/2C/2D/3A/3B)
- **AD:** 20 genes (MAPT, GSK3B, LAMP1, CD33, BCL2, APP, PSEN1/2, APOE, TREM2, CLU, CR1, BIN1, PICALM, ABCA7, MS4A6A, EPHA1, CD2AP, SORL1, FERMT2)
- **KRTAP:** 75 genes (all KRTAP* with measurable CDS and UTR)

Functional Categories

Gene symbol prefixes were used for broad functional classification without external databases: ZNF* (ZincFinger, N=515), SLC* (SoluteCarrier, N=385), OR* with ≤ 6 characters (OlfactoryReceptor, N=425), KRT* (Keratin, N=120), RPL/RPS/MRPL/MRPS (Ribosomal, N=158), GPCR-related prefixes (N=330), HIST/H1/H2A/H2B/H3/H4 (Histone, N=77), HLA-* (N=21), and a curated list of transcription factor prefixes (FOX, SOX, PAX, TBX, HOX, NKX, LHX, GATA, STAT, SMAD, NFkB, TCF, N=208).

Dinucleotide-Preserving Null

For each top outlier gene, the CDS sequence was shuffled 10 times using a dinucleotide-preserving algorithm (preserving all 16 dinucleotide frequencies while randomizing higher-order structure). SHP metrics were recomputed on each shuffled sequence; the mean shuffled fixed_wit served as the null expectation.

Statistical Analysis

Bootstrap confidence intervals: 500 resamples with equal-N=12 subsampling per category pair. Nearest-centroid classification: leave-one-out with Euclidean distance in (CDS fixed_wit, UTR fixed_wit) space. Label permutation test: 200 shuffles of category labels, recomputing LOO accuracy for each shuffle.

Data and Code Availability

Standalone SHP code, demo FASTA input, calibration summaries, figure assets, and preprint materials are provided in the SHP repository:

<https://github.com/JackeyLGene/SHP>. The core SHP implementation and command-line scanner are implemented in Python 3.12 using only the standard library (hashlib, random, math, gzip, csv, json). Figure-generation scripts may require standard scientific plotting packages such as matplotlib and numpy.

CDS and cDNA FASTA files were obtained from Ensembl release 115 (human) and Ensembl Genomes release 62 (non-human species). GENCODE v49 GTF was used for chromosome coordinate mapping (Mudge et al., 2025).

The per-gene SHP matrix (19,491 genes: 8 primary SHP features + 2 gradients + 5 metadata columns) and per-isoform matrix (224,518 transcripts: 4 SHP features + 3 metadata columns) are provided as supplementary CSV files.

Questions, replication reports, biological interpretation notes, and requests for additional matrix files can be sent to jackey.l.gene@outlook.com.

References

- Blum M, Andreeva A, Florentino LC, et al. InterPro: the protein sequence classification resource in 2025. *Nucleic Acids Research*. 2025;53(D1):D444-D456.
- Dyer SC, Austine-Orimoloye O, Azov AG, et al. Ensembl 2025. *Nucleic Acids Research*. 2025;53(D1):D948-D957.
- Gene Ontology Consortium. The Gene Ontology knowledgebase in 2023. *Genetics*. 2023;224(1):iyad031.
- Jaccard P. Etude comparative de la distribution florale dans une portion des Alpes et du Jura. *Bulletin de la Societe Vaudoise des Sciences Naturelles*. 1901;37:547-579.

- Mudge JM, Carbonell-Sala S, Diekhans M, et al. GENCODE 2025: reference gene annotation for human and mouse. *Nucleic Acids Research*. 2025;53(D1):D966-D975.
- Pollard KS, Hubisz MJ, Rosenbloom KR, Siepel A. Detection of nonneutral substitution rates on mammalian phylogenies. *Genome Research*. 2010;20(1):110-121.
- Saussure F de. *Course in General Linguistics*. Edited by Bally C and Sechehaye A. Translated by Harris R. Open Court; 1983.
- Sharp PM, Li WH. The codon adaptation index—a measure of directional synonymous codon usage bias, and its potential applications. *Nucleic Acids Research*. 1987;15(3):1281-1295.
-

Acknowledgements

This work emerged from a broader computational research program on symbolic sequence processing and frame-economy dynamics. The SHP encoding inherits its dual-axis design from PhasePrompt’s chroma-rhythm NLP encoder. The calibration methodology builds on the fair-baseline measurement framework developed in earlier work. The author thanks the developers of Ensembl and GENCODE for maintaining open genomic data resources.